QGIS Basics

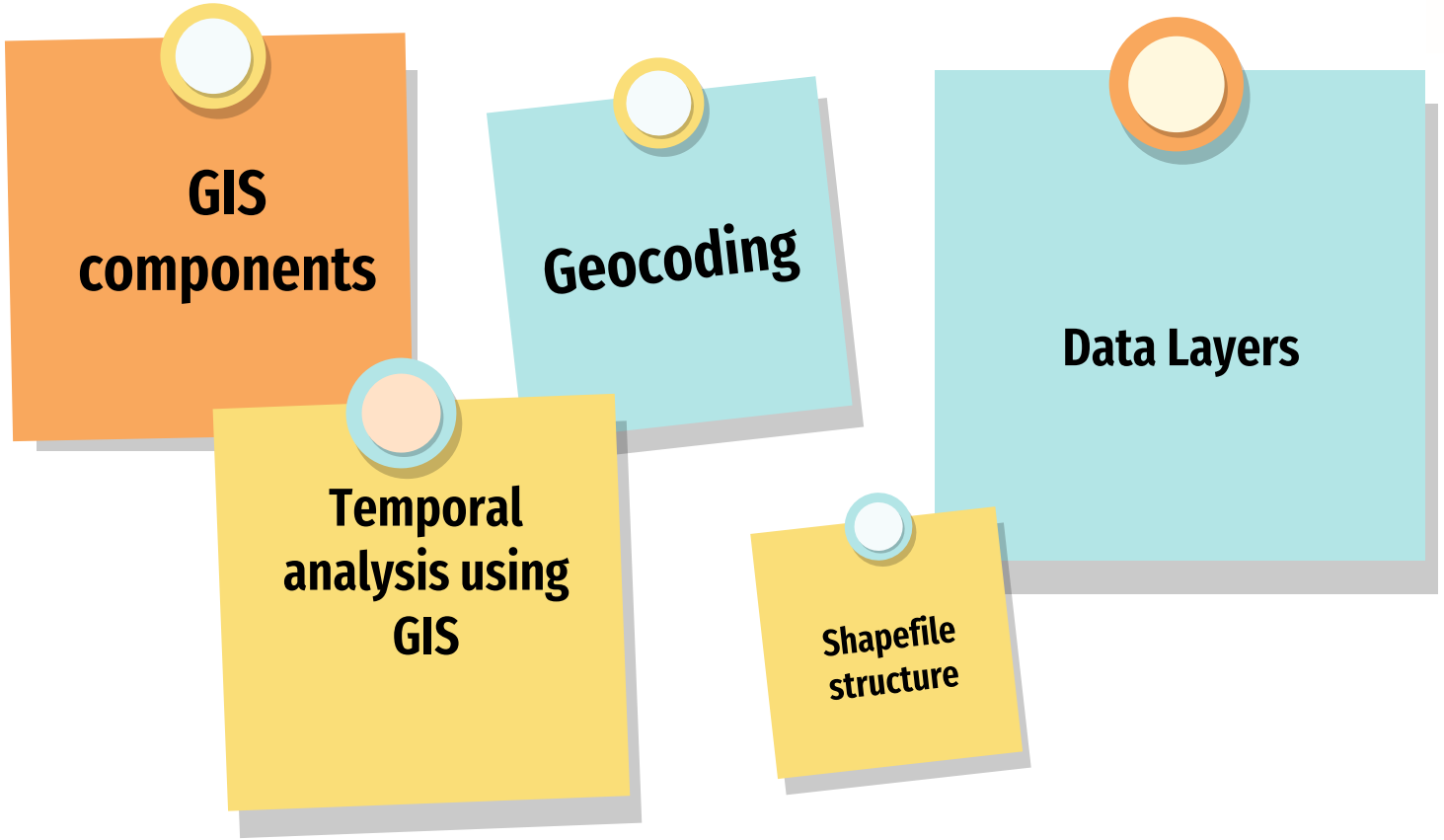
Lab_02: Georeferencing a Historical Map

402.467 / Applied GIS Fundamentals

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21 October 2026

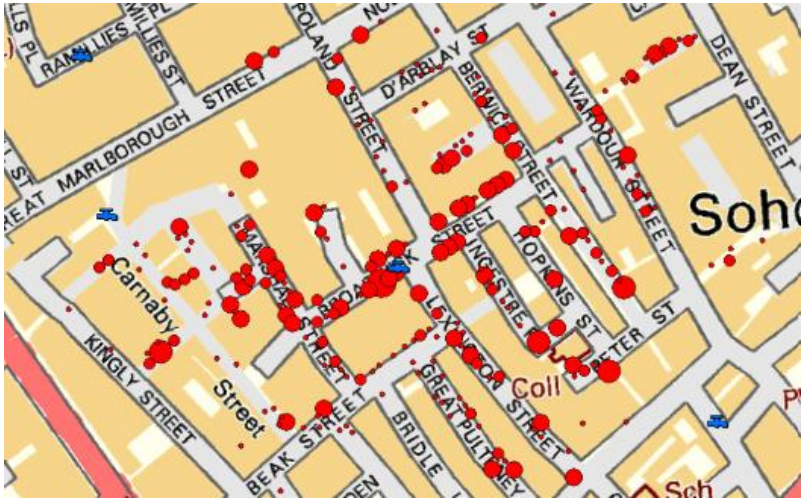
5' Recap



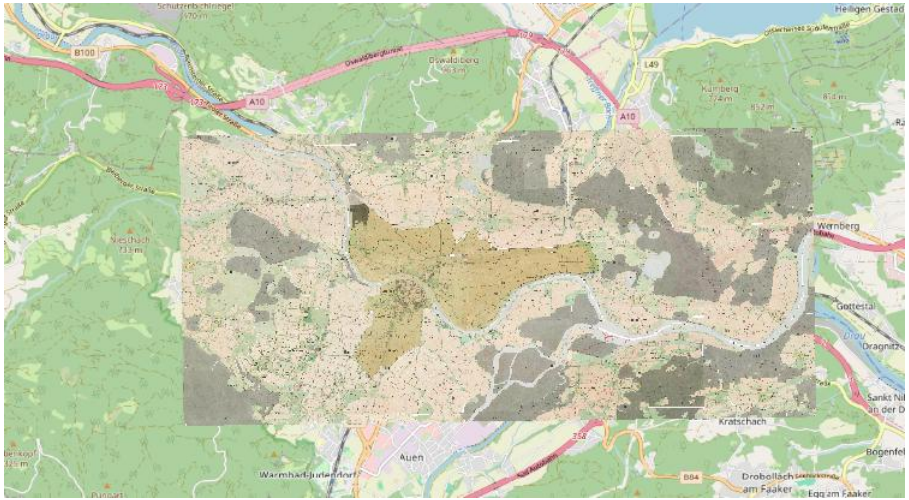
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Georeferencing vs Geocoding

Geocoding: Matching addresses to geographic coordinates (latitude & longitude)



Georeferencing: Matching geographic images to coordinates



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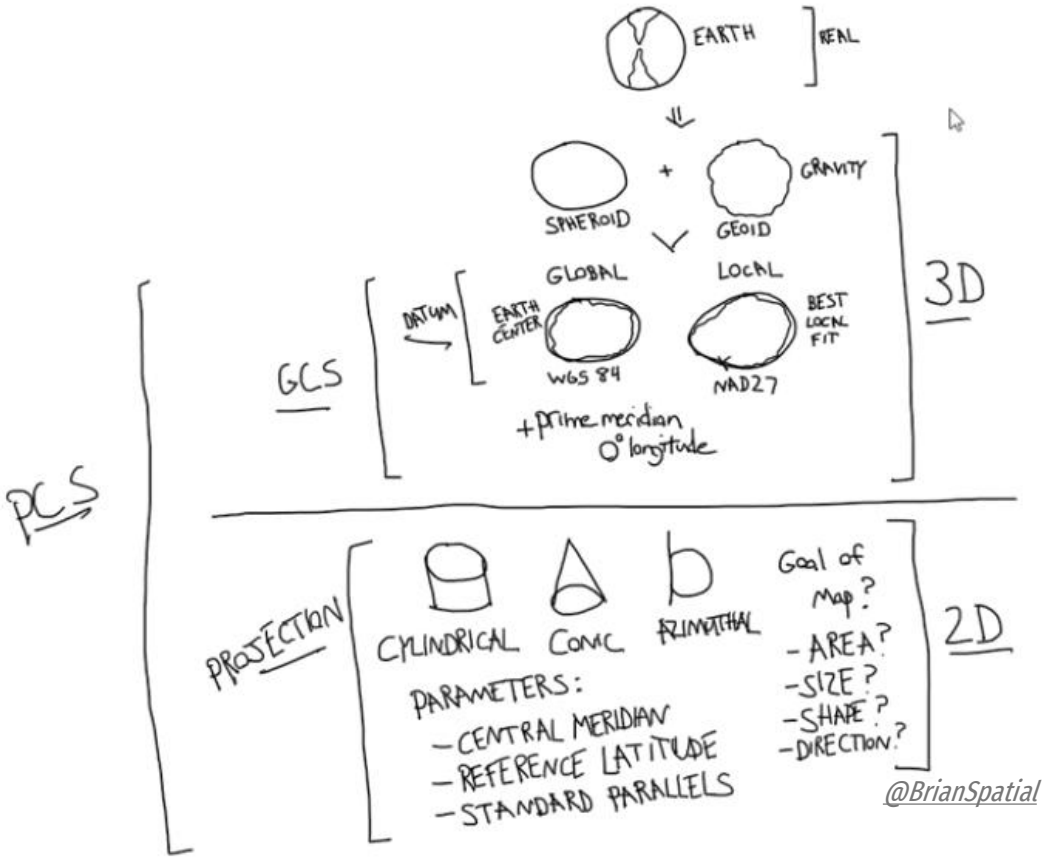
Coordinate Systems

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Introduction

Challenge 🎯

To go from 3D object (Earth) to a 2D representation (a map)

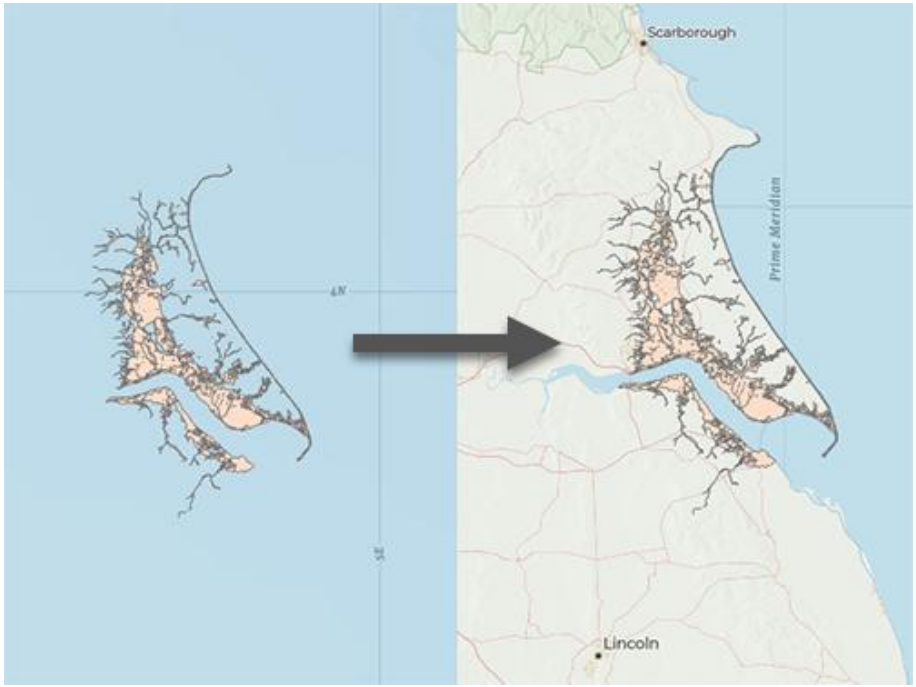


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Coordinate systems

-Q: Why do we need a coordinate system? 🤔

-A: Without a common coordinate reference, GIS data layers won't line up properly! 💡



@ESRI

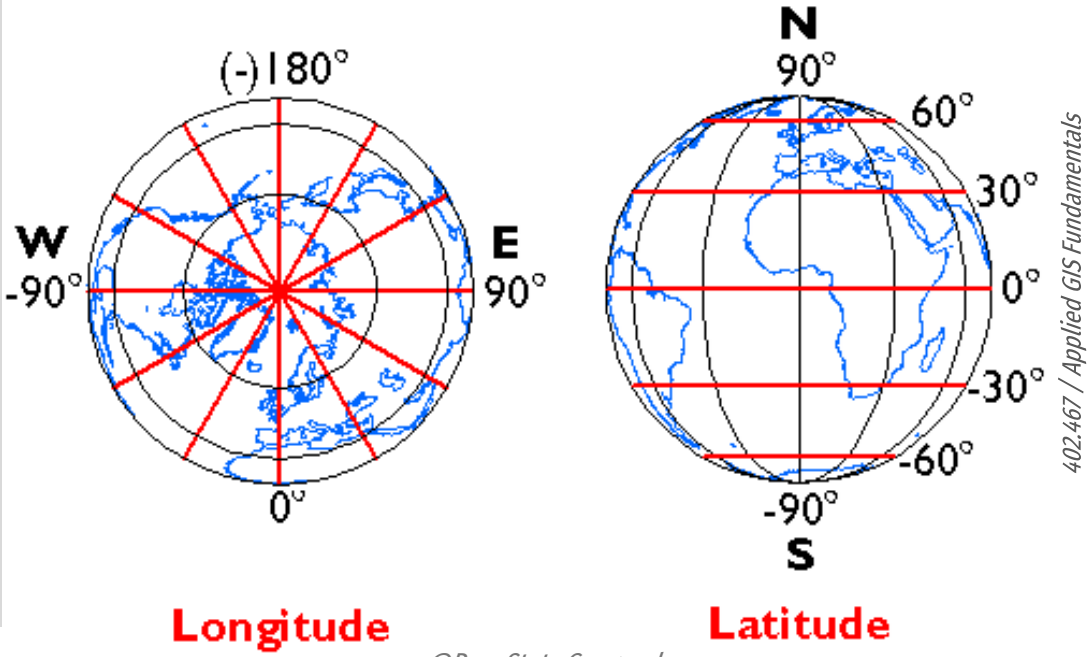
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Defining key terms

Geographic Coordinate System (GCS)

- Identifies locations on the **curved surface** of the Earth
- Locations are defined by two values:

Latitude and Longitude



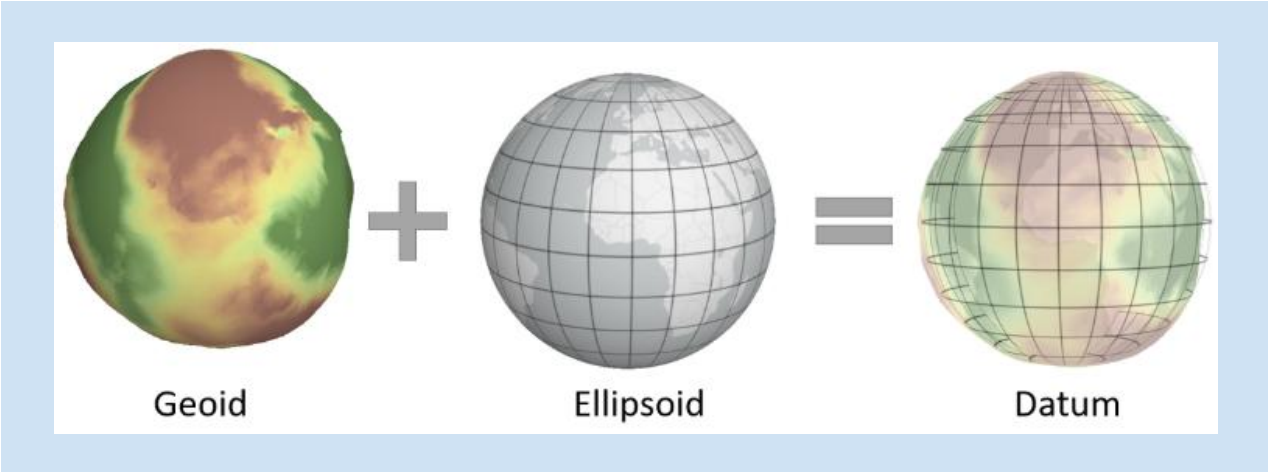
@PennState Geography

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Defining key terms

A GCS is defined by three components

1. Geoid **2. Ellipsoid** **3. Datum**



@Manuel Gimond

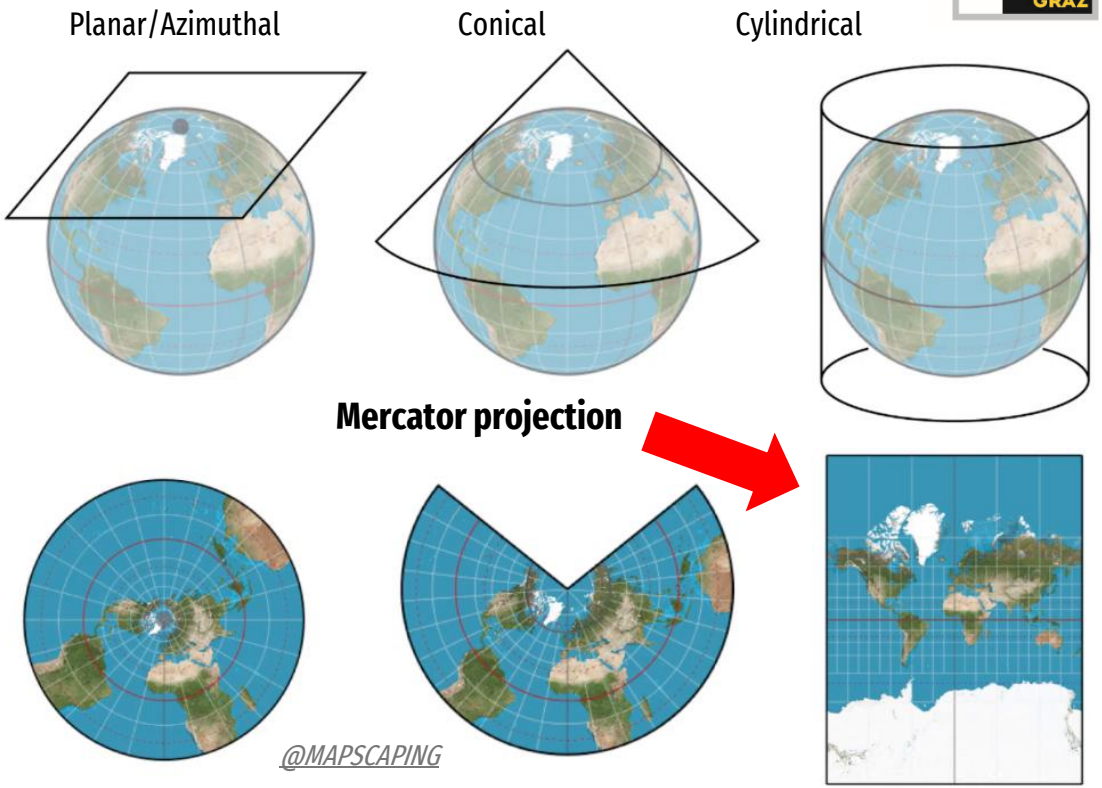
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Defining key terms

Projected Coordinate System (PCS)

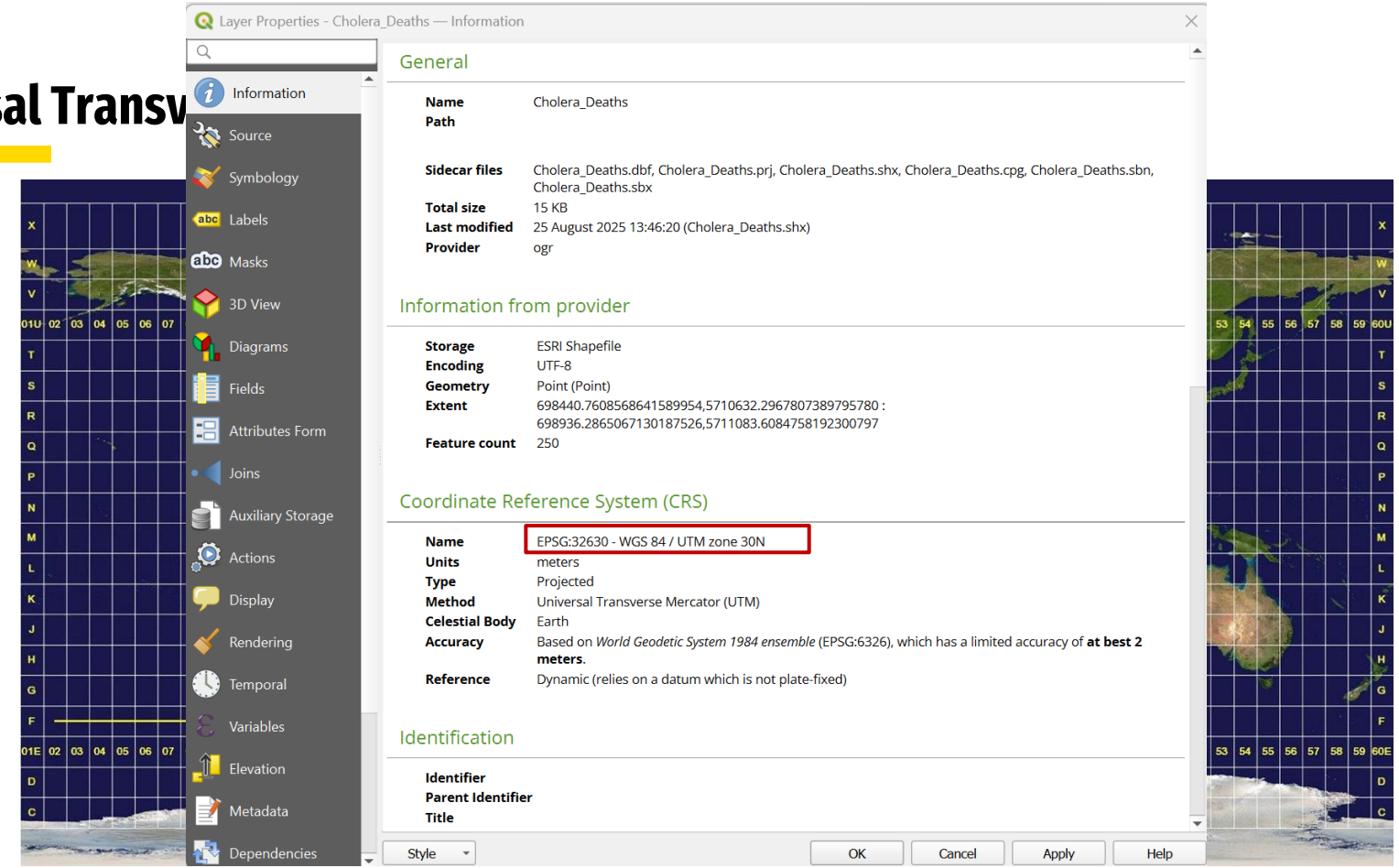
- Identifies locations & measuring features on a **flat surface**
- Transforming data from GCS to PCS requires mathematical projection. There are three broad categories:

Planar/Azimuthal Conical Cylindrical



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Universal Transv



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The orange peel problem in Cartography



@Map projections

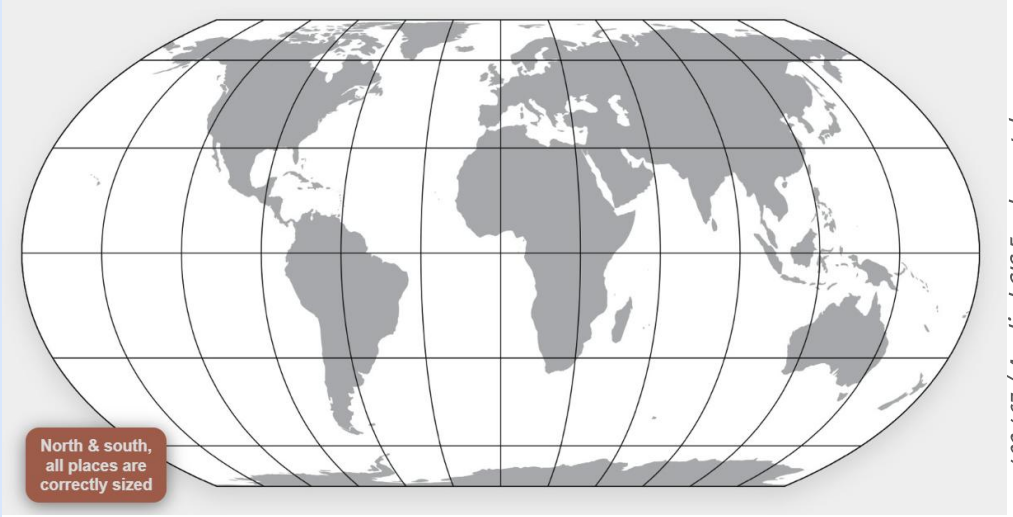
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Something changed on September 4, 2026...

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Equal Earth Projection

- **Moving** from Mercator **to** the **Equal Earth** projection
- Decision from the **UN 164–1 vote**
- Equal Earth **fairly displays relative continent sizes** (e.g., Africa)
- **Digital navigation** infrastructure remains tied to Mercator



@Equal Earth

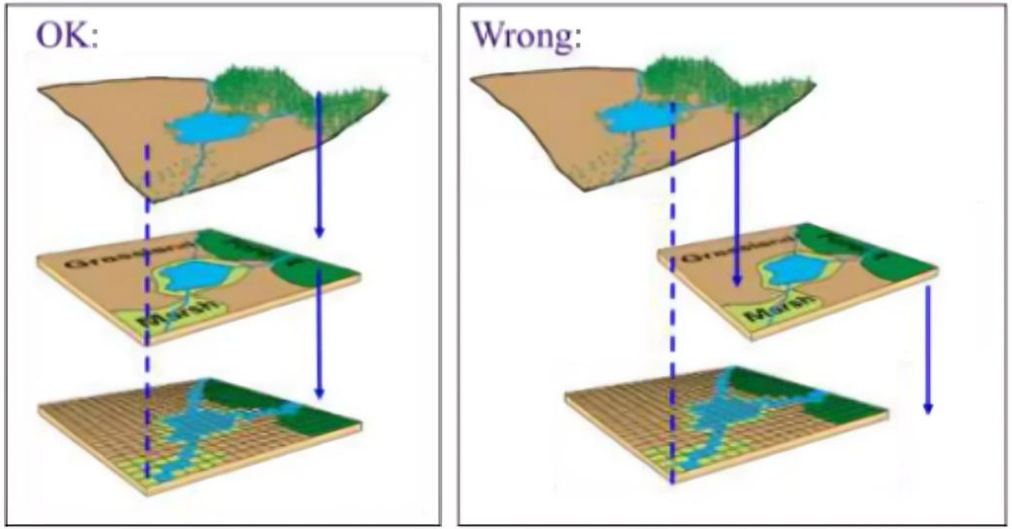
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How does QGIS handle CRS?

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On the fly reprojection

- All layers visualised in a GIS application need to be in the same projection
- Instead of reprojecting all layers to the same projection, GIS applications use '*on the fly reprojection*'



@Hans van der Kwast

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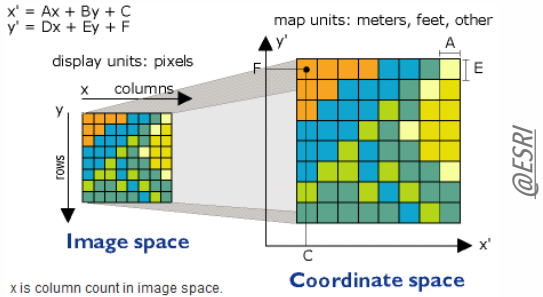
Georeferencing

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Georeferencing

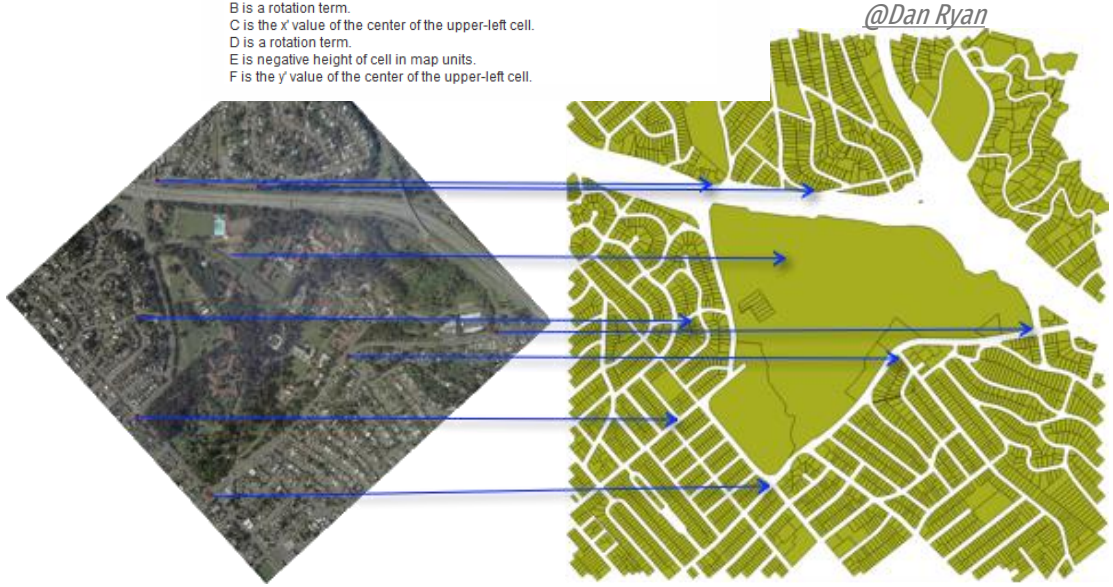
~To georeference means to associate something (e.g., images, maps) with locations to the physical space

~Georeferencing is the process of associating a physical map or raster image of a map with spatial locations



x is column count in image space.
y is row count in image space.
x' is horizontal value in coodinate space.
y' is vertical value in coordinate space.

A is width of cell in map units.
B is a rotation term.
C is the x' value of the center of the upper-left cell.
D is a rotation term.
E is negative height of cell in map units.
F is the y' value of the center of the upper-left cell.



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Georeferencing raster images

~Register a map with the correct real world coordinates

~If the maps are not georeferenced, no other information can be displayed over or positioned under the map coverage

~Depending on the source materials & specifications of the project, maps can be registered to either a coordinate system or to another basemap such as digital orthophoto



@OpenDroneMap

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Entering GCPs

Ground Control Points are the foundation for accurately georeferencing a raster image

They are known locations on the Earth's surface and are identified by their own latitude & longitude

💡 Use multiple GCPs across an area & distribute them evenly for better accuracy

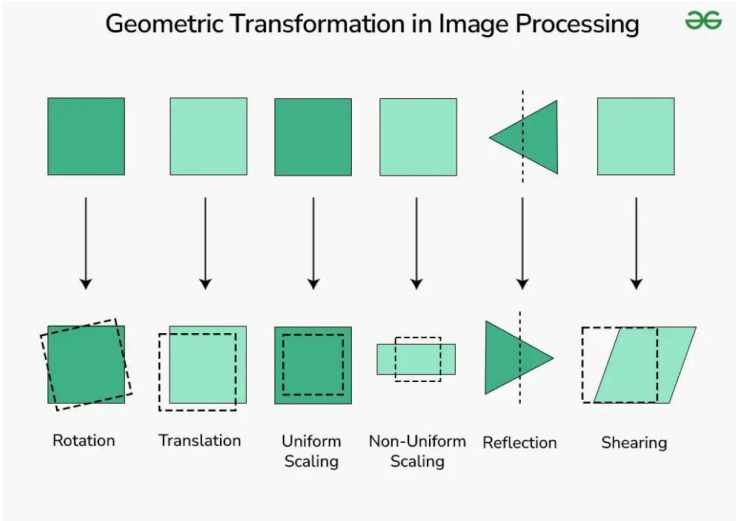


GCP	Pixel X	Pixel Y	Longitude	Latitude
1	4203.7	2347.0	4.2946	52.0825
2	3830.5	1673.9	4.2744	52.0888
3	5122.7	1611.8	4.3054	52.1025
4	5522.5	2981.5	4.3371	52.0862
5	3593.9	2691.5	4.2849	52.0710
6	6560.4	4341.7	4.3844	52.0761
7	4432.0	4929.8	4.3406	52.0464
8	486.8	2458.7	4.2042	52.0447
9	1171.9	4210.5	4.2483	52.0252
10	6704.4	907.2	4.3332	52.1289
11	1880.0	1993.3	4.2314	52.0650

@kokoalberti

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Transformation types

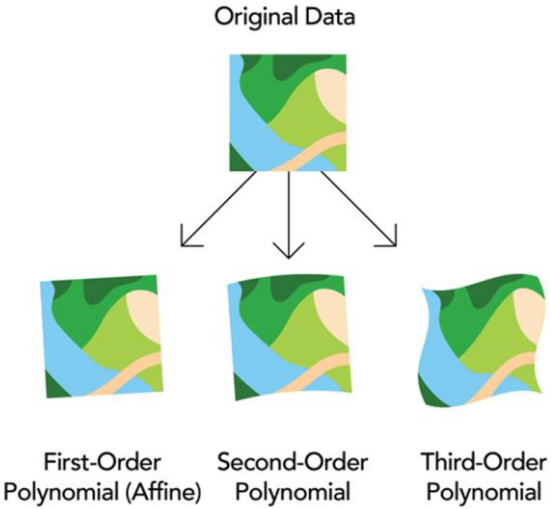


@Geekfofgeeks

GCPs are used to create a **mathematical equation** for aligning raster data to real-world coordinates

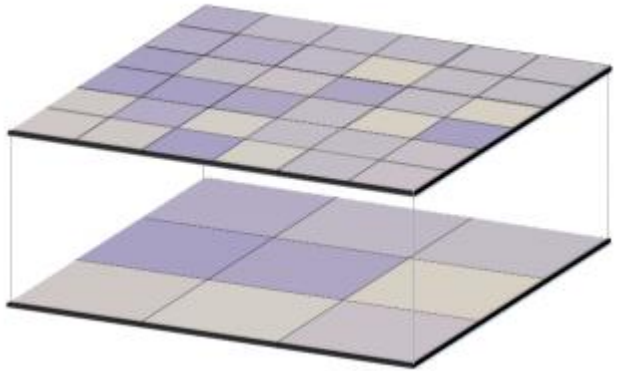
Most common method: **affine transformations** (first-order polynomial = linear), allows

- **Translation** (shifting)
- **Scaling** (resizing)
- **Rotation** (turning)
- **Shearing** (skewing)



@ESRI

Resampling method



- Nearest Neighbor
- Bilinear
- Cubic Convolution
- Majority

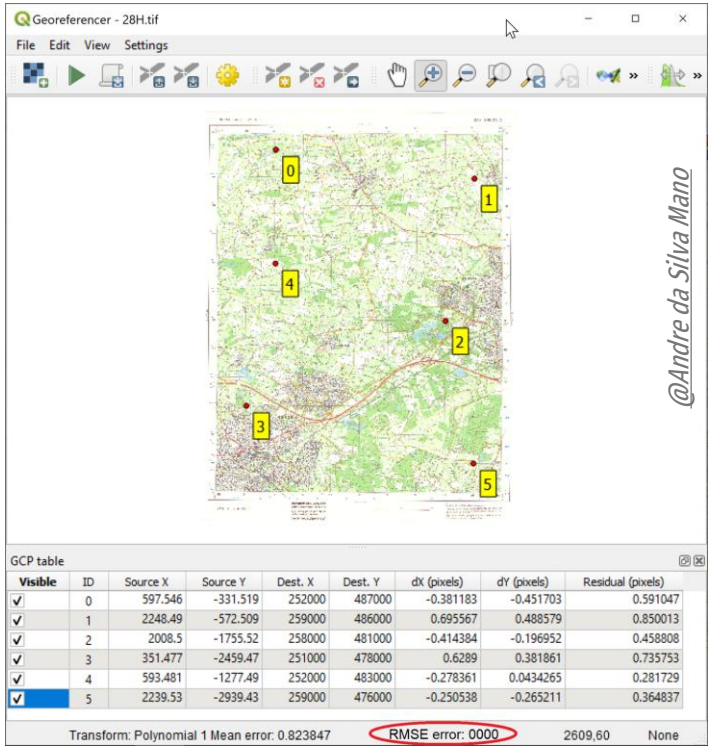
~Choosing the correct resampling method determines how new pixel values are calculated for the output image

~Each method has different effects on the output image's appearance, accuracy and processing time

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RMSE

- **What is the Root Mean Square Error (RMSE) ?**
- **It is a statistical measure of the difference between the GCPs & their true more accurate coordinates**



$$RMSE = \sqrt{\frac{\sum_{i=1}^{n}(P_i - O_i)^2}{n}}$$

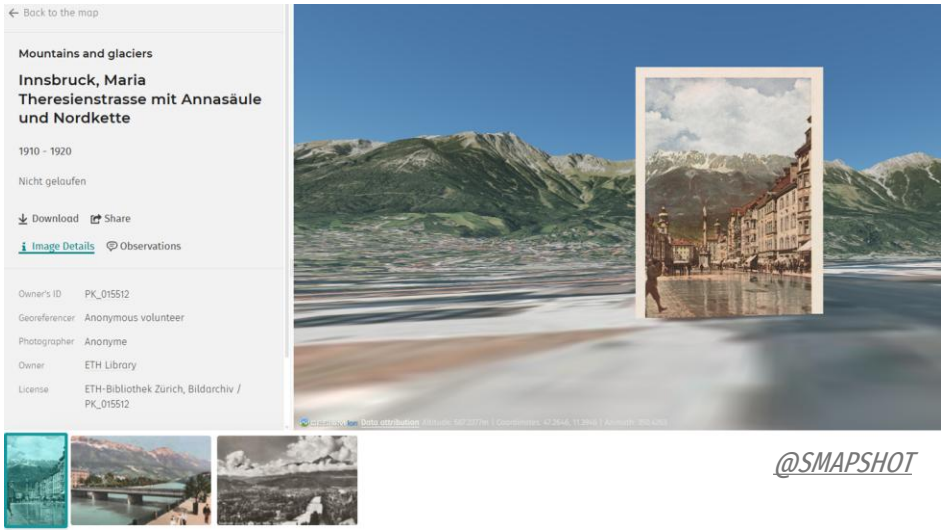
@GISGeography

A **lower RMSE** indicates a **better fit** / more accurate georeferencing. A **high RMSE** suggests **poor GCP selection**, an **inappropriate transformation model**, or **inherent distortions** in the source raster

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Examples

Smapshot: The participative time machine



@SMAPSHOT

Map Warper



@Map Warper